MILITARY SPECIFICATION

MAINTENANCE KIT, FIBER OPTIC COMPONENTS, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for a maintenance kit used for repair and maintenance of fiber optic components.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-C-1752	-	Cushioning Material, Packaging, Unicellular Polyethylene Foam, Flexible.

MILITARY

MIL-C-4150	-	Case, Transit and Storage, Waterproof and Water-Vaporproof.
MIL-P-19834	-	Plate, Identification, Metal Foil, Adhesive Backed.
MIL-C-22520	-	Crimping Tools, Terminal, Hand or Power Actuated, Wire Termination, and Tool Kits.
MIL-T-45542	-	Tool Sets, Shop Sets, and Kits (Hardware, Installation, Modification, and Tool), Packaging of.
MIL-F-81334	-	Foam, Plastic, Flexible, Open Cell, Polyester Type Polyurethane.
MIL-I-81969	-	Installing and Removal Tools, Connector Electrical Contact, General Specification for.
MIL-T-83523	-	Tools, Fiber Optic.
MIL-K-83525	-	Kit, Portable Optical Microscope, Militarized, 200X Magnification for Field Inspection of Optical Fibers.
MIL-K-83533/1	-	Maintenance Kits, Fiber Optic Connectors, Contacts, and Connector and Cable Accessories: Maintenance Kit, Fiber Optic Connector Terminations.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Electronic Support Division AFLC, 2750 ABW/ES, Gentile AF Station, Dayton, Ohio 45444-4500, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6080

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STANDARDS

FEDERAL

FED-STD-595 - Color.

MILITARY

MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Materials. Materials shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the kits to meet the performance requirements of this specification and the applicable specification sheet. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.3.1 Cases. Unless otherwise specified (see 3.1), the case shall be of the following categories.

3.3.1.1 Category 1. Ruggedized case shall meet the requirements of MIL-C-4150.

3.3.1.2 Category 2. Environmental case shall not be degraded by its environment (see 4.3.2).

3.3.1.2.1 Low pressure. When tested as specified in 4.7.2 cases will not show permanent dimensional deformation.

3.3.1.2.2 High temperature. When tested as specified in 4.7.3 cases shall open, close, latch and seal without binding. Case materials shall not deform or separate.

3.3.1.2.3 Low temperature. When tested as specified in 4.7.4 cases shall not crack, deform or bind.

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3.3.1.2.4 Solar radiation. When tested as specified in 4.7.5 using the full solar spectrum there shall be no evidence of active induced degredation such as material separation or peeling.

3.3.1.2.5 Humidity. When tested as specified in 4.7.6 case materials shall not exhibit swelling.

3.3.1.2.6 Fungus. When tested as specified in 4.7.7 cleaned cases shall not support the growth of fungus.

3.3.1.2.7 Salt spray. When tested as specified in 4.7.8 there shall be no corrosion visible to the unaided eye.

3.3.1.2.8 Sand and dust. When tested as specified in 4.7.9 there shall be no dust or sand inside the case at the completion of the test.

3.3.1.2.9 Leakage. When tested as specified in 4.7.10 there shall be no evidence of leakage to the interior of the case.

3.3.1.2.10 Vibration. When tested as specified in 4.7.11 the case shall protect intended contents from any vibration damage.

3.3.1.2.11 Shock. When tested as specified in 4.7.12 the case shall protect intended contents from shock damage.

3.3.1.3 Category 3. Case shall be nonenvironmental.

3.3.2 Cushioning material. Unless otherwise specified (see 3.1), cushioning material shall be in accordance with PPP-C-1752 or MIL-F-81334.

3.4 Design and construction. Kits shall be of the design, construction, and physical dimensions specified (see 3.1).

3.4.1 Cases. Cases, metal or fiberglass shall be designed as specified (see 3.1).

3.4.2 Color. The kit color shall be in accordance with FED-STD-595.

3.4.3 Installing and removal tools. Unless otherwise specified (see 3.1), all installing and removal tools shall be in accordance with MIL-I-81969, or approved equivalent.

3.4.4 Crimping tools, dies, turrets, positioners, and gauges. Unless otherwise specified (see 3.1), all crimping tools, dies, turrets, positioners, and gauges shall be in accordance with MIL-C-22520, or approved equivalent.

3.4.5 Optical microscope. Unless otherwise specified (see 3.1), all optical microscopes shall be in accordance with MIL-K-83525, or approved equivalent.

3.4.6 Epoxies. Use Trabond 230R, Eccobond 144BR, or an equivalent epoxy approved by the qualifying activity (see 6.4).

3.4.7 Repair parts. Bulk repair parts such as splices, termini, connectors and cable described by military specification or military adopted specification are preferred when included as part of the kit.

3.4.8 Fiber optic tools. Unless otherwise specified (see 3.1), all fiber optic tools shall be in accordance with MIL-T-83523, or approved equivalent.

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3.5 Marking. All marking shall be in accordance with MIL-STD-130 and shall be permanent and legible.

3.5.1 Nameplates. Nameplates on tool cases shall be marked with information as specified (see 3.1). The plates shall be 0.003 to 0.005 inch (0.08 to 0.13 mm) thick aluminum foil in accordance with MIL-P-19834 and shall be permanently mounted. Marking shall be black on a light background.

3.5.2 Content charts and instruction sheets. Content charts and instruction sheets shall be white vinyl 0.020 inch (0.51 mm) thick or approved equivalent with black lettering. The lettering shall be protected with a transparent coating.

3.6 Workmanship. The tool cases and tools contained therein shall have no sharp edges, burrs, or other features which are potential personal hazards to the operator during normal usage.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The inspection of the kit shall be classified as follows:

a. First article inspection (see 4.3).

b. Materials inspection (see 4.4).

c. Quality conformance inspection (see 4.6).

4.3 First article inspection. First article inspection shall be performed by the contractor after award of contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production. First article approval is valid only on the contract or purchase order under which it is granted, unless extended by the Government to other contracts or purchase orders.

4.3.1 Sample size. Four samples of each PIN shall be subjected to first article inspection to assure that cases are able to function in severe environments.

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4.3.2 Inspection routine. The sample shall be subjected to the inspections specified in table I, in the order shown (see 3.3.1.2).

TABLE I. Inspection routine.

Inspection	Requirement paragraph	Method paragraph
Low pressure	3.3.1.2.1	4.7.2
High temperature	3.3.1.2.2	4.7.3
Low temperature	3.3.1.2.3	4.7.4
Solar radiation	3.3.1.2.4	4.7.5
Humidity	3.3.1.2.5	4.7.6
Fungus	3.3.1.2.6	4.7.7
Salt spray	3.3.1.2.7	4.7.8
Sand and dust	3.3.1.2.8	4.7.9
Leakage	3.3.1.2.9	4.7.10
Vibration	3.3.1.2.10	4.7.11
Shock	3.3.1.2.11	4.7.12

4.3.3 Failures. One or more failures shall be cause for refusal to grant approval.

4.4 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials, as specified herein and on the specification sheet (see 3.1), used in fabricating the kits, are in accordance with the applicable referenced specifications or requirements prior to such fabrication.

4.5 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in MIL-STD-810.

4.6 Quality conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A inspection.

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4.6.1.1 Group A inspection. Group A inspection shall consist of the inspections specified in table II, in the order shown. All units shall be tested and no failures shall be accepted.

TABLE II. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph	Allowable defects	
			Major	Minor
Visual and mechanical inspection:				
Material	3.1, 3.3	4.7.1	0	0
Finish	3.1, 3.3	4.7.1	0	0
Design and construction	3.4	4.7.1	0	0
Marking	3.5	4.7.1	0	0
Workmanship	3.6	4.7.1	0	0

4.6.1.1.1 Disposition of group A sample. The kit which has passed group A inspection may be delivered on the contract.

4.6.2 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of MIL-T-45542.

4.7 Methods of inspection.

4.7.1 Visual and mechanical inspection. The kit and contents shall be examined to verify that the design, construction, physical dimensions, markings, and workmanship are in accordance with applicable requirements (see 3.1, 3.3, 3.4, 3.5, and 3.6).

4.7.2 Low pressure (see 3.3.1.2.1). Cases shall be tested in accordance with MIL-STD-810 method 500, procedure II and III. Test altitude shall be 12,200 meters using rapid decompression rate. Test duration to be no less than 1 hour.

4.7.3 High temperature (see 3.3.1.2.2). Cases shall be tested in accordance with MIL-STD-810 method 501, procedure II. Test duration 24 hours performed at +71°C. A minimum of 3 cycles must be performed.

4.7.4 Low temperature (see 3.3.1.2.3). Cases shall be tested in accordance with MIL-STD-810 method 502, procedure II. Test duration 72 hours at a temperature of -51°C.

4.7.5 Solar radiation (see 3.3.1.2.4). Cases shall be tested in accordance with MIL-STD-810 method 505, procedure II. Tests will be for 10 cycles, 20 hours each at 355 BTU/ft^2/h.

4.7.6 Humidity (see 3.3.1.2.5). Cases shall be tested in accordance with MIL-STD-810 method 507, procedure II, for nonhazardous test items.

4.7.7 Fungus (see 3.3.1.2.6). Cases shall be tested in accordance with ASTM G20-70.

4.7.8 Salt spray (see 3.3.1.2.7). Cases shall be tested in accordance with MIL-STD-810 method 509, duration shall be 48 hours continuous exposure.

4.7.9 Sand and dust (see 3.3.1.2.8). Case shall be tested in accordance with MIL-STD-810 method 510. Case shall be closed and latched.

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4.7.10 Leakage (see 3.3.1.2.9). Case shall be tested in accordance with MIL-STD-810 method 512.

4.7.11 Vibration (see 3.3.1.2.10). Case shall be tested in accordance with MIL-STD-810 method 514, category 1.

4.7.12 Shock (see 3.3.1.2.11). Cases shall be tested in accordance with MIL-STD-810, method 516, procedure 1.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-T-45542 for type I kits. In addition, unit packs and shipping containers shall be marked with the following information:

a. Number (revision and amendment, if applicable) of this commodity specification.

b. PIN and title.

c. Manufacturer's name.

d. Manufacturer's part number.

e. Month and year inspected.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Kits and contents therein covered by this specification are intended to be used for repair and maintenance of fiber optic cable, connectors, termini, modules, and other fiber optic components.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Part or Identifying Number of individual components included in kits shall be specified.

d. Part or Identifying Number for maintenance kits as required.

e. Part or Identifying Number for replacement parts for the kit when ordered separately.

6.3 First article. When a first article inspection is required, the items should be a sample selected from the first four production items. The first article should consist of four units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

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6.4 Trademarks. Trabond 230R is a trademark of TRA-CON, Incorporated. Eccobond 144BR is a trademark of Grace WR and Company, Emerson and Cuming Dewey and Almy Chemical Division.

6.5 Part or Identifying Number (PIN). The PIN should be as specified on applicable specification sheet (see 3.1).

6.6 Subject term (key word) listing.

Tools, adapter, container, case, channel wrench.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, AT, GL, MI
Navy - AS, OS, SH,
Air Force - 11, 12, 13, 14, 17, 19, 80, 99
DLA - ES

User activities:
Army - AV, ME
Navy - MC
Air Force - 70, 82, 84, 90

Preparing activity:
Air Force - 85

Agent:
DLA - ES

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